

29 – M9 Integration

Module design for the outward surface: the public API as a versioned OpenAPI contract with scoped OAuth2 clients, journal export derived from the posting maps for the accounting system to come, notification rules and templates with provider adapters, the e-invoicing seam, and the webhook design held for phase 2. Adapters translate; they never hold a business rule.

Field	Value
Document	29 – M9 Integration (Stage 2 module design; template of doc 20 §5)
Version	v0.1 – draft for design review; contract freeze targeted end of Stage 2 week 7 with doc 33
Date	9 September 2026
Owning developer	Module developer (full-stack); the API bundle (doc 33) is assembled here from all module slices
Inputs	05 SRS (FR-BIL-110, FR-BIL-140); 16 answers (J-02, J-04, R-05 to R-09, M-10); 09 ADR-01, ADR-05; 17 v0.2.1 §7.9, §10; 18 v0.2; 19 v0.2 §7 (notifications delivery), §8; 20 v0.3 M9; 21–28 contracts (events, posting maps, notification rules handed over); 30 §5.9; 32
Depends on	All modules' queries and events; M1 (client scope, entities), M4 (posting maps, invoice snapshots), M5 and M7 (postings), kernel (identity, notifications delivery, attachments, scheduling, audit)
Depended on by	The accounting system (first consumer), the Inland Revenue adapter (phase 3), SMS and e-mail providers, doc 33

Two principles. An adapter is a translator, never a place for business rules: the posting map is M4's data, the invoice snapshot is M4's document, the notification rule fires on another module's event – M9 only carries them outward in the shape the outside world needs (doc 17 §10.2). And the public API exposes module commands and queries under the same scope, permission and idempotency rules as the back office: a client is a principal, not a back door.

1. Purpose and scope

In scope (phase 1)	Out of scope
API clients: registration per entity, OAuth2 client credentials, scoped grants, rotation, suspension, rate limits, call audit (R-07)	Inbound write APIs beyond what the accounting consumer needs (acknowledgements); a public developer programme
The public OpenAPI surface: read endpoints over documents, stock, exposure, statements, journal exports, scoped by the client's entity and the same RLS classes; versioning and deprecation policy; doc 33 assembled here	GraphQL; an API gateway product (doc 20 §4)
Journal export: per entity and period, lines derived from the posting maps of every mapped document type, generic JSON and CSV, immutable once generated, acknowledged by the consumer (J-02, FR-BIL-140)	Statutory accounting of any kind (ADR-05); a specific accounting package's format until it is chosen (adapter slot)
Notification rules, templates (three languages) and provider adapters for SMS and e-mail; in-app channel; quiet hours and kill switch configuration (R-08)	The delivery mechanics (kernel, doc 19 §7)
E-invoicing seam: submission record, adapter interface, stub implementation; readiness only (J-04, FR-BIL-110)	Live Inland Revenue integration (phase 3)
Webhook subscriptions designed for phase 2 on the consumer mechanism	Built in phase 1
Regulator read-only access (M-10) as an M1 grant with M8 exports	Prescribed return formats (none exist)

2. Requirements coverage

Identifier	Requirement (abridged)	Design section
FR-BIL-110	E-invoicing readiness: immutable invoice snapshot, reserved integration point	3.5, 4.4
FR-BIL-140	Postings derived from documents through the posting map	3.3
J-02	Export format decided when the accounting system is chosen; mapping exists now	3.3, 11 (DR-2)
J-04	E-invoicing expected; readiness built in	3.5
R-05	No bank integration	1 (out of scope)
R-06	No government portal now; readiness	3.5
R-07	OpenAPI, OAuth2/OIDC-secured; accounting system first consumer	3.1, 3.2, 4.1
R-08	SMS and e-mail notifications: dispatch alerts, statements, member communications	3.4
R-09	No inherited devices	–
M-10	Department of Co-operative Development read-only access	3.6
ADR-01	Money recorded, never moved	3.3 (export only)
Doc 19 §7	Kernel delivers; M9 owns rules, templates, providers	3.4

3. Domain model

Schema integration.

3.1 API clients

Aspect	Design
Aggregate	<code>api_client(client_id, owner_entity_id, name, kind ACCOUNTING THIRD_PARTY INTERNAL_TOOL, credential_kind SECRET CERT, secret_hash, cert_subject, scopes[], rate_limit_per_minute, status, created_by)</code> plus <code>api_credential_history</code> for rotation (old credential valid for an overlap window).
Principal	A client authenticates with OAuth2 client credentials at the identity provider (doc 19 §2) and receives a token with <code>cls = API_CLIENT</code> , <code>ent = its owner entity</code> , and its scopes. The kernel scope filter maps it to <code>OWN</code> for its entity (or <code>PARTY</code> on documents where its entity is a party); it never gets <code>FEDERATION_VIEW</code> unless the Federation registers it so.
Scopes	Named API scopes (<code>documents:read</code> , <code>stock:read</code> , <code>exposure:read</code> , <code>statements:read</code> , <code>journal:read</code> , <code>journal:ack</code> , <code>customers:read</code> (MPCS only)); each maps to a permission set; granted by the entity's administrator within their own permissions (no escalation, doc 19 §3).
Call audit	<code>api_call_log(client, at, operation, status, latency, rows)</code> sampled at 100% for writes and 10% for reads; rate limiting per client at the edge and in the interceptor; 429 with <code>retry-after</code> .

3.2 The public API

Aspect	Design
Shape	REST under <code>/v1</code> (doc 17 §10.1): OpenAPI 3.1, bearer tokens, idempotency keys on mutations, cursor pagination, problem details, ETags on documents. The bundle (doc 33) is assembled from the module slices; the public subset is marked <code>x-public: true</code> and is the only part clients see.
Phase-1 public operations	Read: invoices, credit and debit notes, payment receipts (seller and buyer views through <code>PARTY</code> masking), GRNs (own), stock balances (own), exposure and statements (own relationship side), journal exports (own), customer accounts and postings (own MPCS, <code>customers:read</code>). Write: acknowledge a journal export; nothing else in phase 1.
Versioning	<code>/v1</code> is stable for the phase; additive changes only; a breaking change opens <code>/v2</code> with <code>/v1</code> supported for at least 12 months and a <code>Sunset</code> header; deprecations listed in the bundle.

3.3 Journal export

Aspect	Design
Derivation	For an entity and period, <code>JournalBuilder</code> walks every document with a posting map (M4 <code>posting_map: INV, CN, DN2, PRC</code> seller side; GRN and INV buyer side; plus M5 WOF/ADJ/RPK and M7 CPR maps registered the same way) and produces <code>journal_line(export, document, line_kind, side, debit_role, credit_role, amount, business_date, reference)</code> with account <i>roles</i> ; the chart-of-accounts mapping is applied by the package adapter when one exists, else roles are exported as-is (J-02).
Export	<code>journal_export(export_id, entity, period_from, period_to, format JSON CSV ADAPTER, status, object_key, line_count, total_debit, total_credit, content_hash, generated_at, acknowledged_by, acknowledged_at, supersedes)</code> . Immutable once <code>GENERATED</code> ; a correction is a new export that supersedes the old (documents issued later for the period appear in a supplementary export, never by editing).
Balance	Every export balances ($\Sigma \text{debit} = \Sigma \text{credit}$) by construction of the posting map; a failure is a defect in a map, reported as an <code>ALERT</code> , never silently exported.
Delivery	Object storage plus the API (<code>journal:read</code>); the consumer acknowledges (<code>journal:ack</code>); unacknowledged exports older than 30 days raise a <code>REVIEW</code> .

3.4 Notifications: rules, templates, providers

Aspect	Design
Rules	<code>notification_rule(rule_id, owner_entity_id (null = federation), event_type, predicate jsonb, template_id, audience_kind ROLE_AT_COUNTERPARTY ROLE_AT_OWNER CUSTOMER EXPLICIT, audience_spec, channels[], priority, status)</code> . Modules hand over their rules as seed data (M4 dispatch and dispute events, M7 statements, M8 exception reports, M1 lockouts); an MPCS may add rules for its own events within its scope. The kernel dispatcher (doc 19 §7) evaluates them.

Aspect	Design
Templates	notification_template(template_id, channel, subject_en/si/ta, body_en/si/ta, placeholders_schema, status); ICU MessageFormat; rendered in the recipient's language with English fallback; SMS bodies limited to 160 GSM or 70 UCS-2 characters with a length check at authoring for Sinhala and Tamil.
Providers	provider_config(channel, primary_provider, secondary_provider, credentials_ref (KMS), sender_id, daily_cap, status); adapters implement NotificationChannel (kernel interface): HttpSmsAdapter (provider-neutral HTTP with a small template per gateway), SmtEmailAdapter, InAppAdapter.
Consent and suppression	Customer channels require M7's STATEMENTS consent (checked at audience resolution); quiet hours, de-duplication and the kill switch are configuration items read by the kernel dispatcher.

3.5 E-invoicing seam

Aspect	Design
Readiness	e_invoice_submission(invoice_document_id, status NOT_SUBMITTED PENDING SUBMITTED ACCEPTED REJECTED, provider_ref, submitted_at, response_key, attempts); EInvoiceAdapter interface with StubAdapter (records PENDING, never sends) in phase 1; the payload is M4's immutable invoice snapshot (content hash, both VAT numbers, tax point); nothing on the invoice changes on submission (J-04).

3.6 Regulator and auditor access

Aspect	Design
Path	Not an API: a time-boxed EXTERNAL grant (M1) to the web back office with M8 exports; every export audited. If the Department later prescribes a return format, it becomes an M8 definition and, if needed, an M9 delivery adapter.

3.7 Webhooks (phase 2 design)

Aspect	Design
Design only	webhook_subscription(client, event_types[], url, secret, status) and webhook_delivery log; a kernel consumer per subscription signs and posts the envelope with retries; entity-scoped events only. Built on the same consumer mechanism; not in phase 1.

4. State machines

4.1 API client

From	To	Trigger / command	Guard	Permission	Audit
–	ACTIVE	RegisterApiClient(name, kind, scopes, credential kind)	entity admin; scopes within grantor's permissions; MFA	int.client.manage (MFA)	API_CLIENT_REGISTERED
ACTIVE	ACTIVE	RotateCredential	overlap window (default 7 days) during which both credentials work	int.client.manage (MFA)	API_CREDENTIAL_ROTATED
ACTIVE	ACTIVE	AmendScopes / SetRateLimit	within grantor's permissions; reason	int.client.manage (MFA)	API_CLIENT_SCOPES_CHANGED
ACTIVE	SUSPENDED	SuspendClient(reason)	reason	int.client.manage	API_CLIENT_SUSPENDED
SUSPENDED	ACTIVE	ReinstateClient	reason	int.client.manage (MFA)	API_CLIENT_REINSTATED
ACTIVE / SUSPENDED	REVOKED	RevokeClient	terminal; tokens rejected immediately	int.client.manage (MFA)	API_CLIENT_REVOKED

4.2 Journal export

From	To	Trigger / command	Guard	Permission	Audit
–	REQUESTED	RequestJournalExport(entity, period, format)	entity accounts; period closed (all locations' day-close past period end) or explicit 'provisional' flag	int.journal.export	JOURNAL_REQUESTED
REQUESTED	GENERATED	JournalBuilder completes	balances; hash computed; object stored	system	JOURNAL_GENERATED
REQUESTED	FAILED	map error	ALERT with the offending document and map	system	JOURNAL_FAILED
GENERATED	ACKNOWLEDGED	AcknowledgeJournalExport (API or web)	consumer client with journal:ack or entity accounts	int.journal.ack	JOURNAL_ACKNOWLEDGED
GENERATED/ACKNOWLEDGED	SUPERSEDED	a later export for an overlapping period	supersedes link	system	JOURNAL_SUPERSEDED

4.3 Notification rule and template

From	To	Trigger / command	Guard	Permission	Audit
–	DRAFT	DefineRule / DefineTemplate	owner scope; event type exists; template placeholders match the event's payload schema; SMS length check	int.notify.manage	NOTIFY_RULE_DEFINED
DRAFT	ACTIVE	ActivateRule	template ACTIVE in all three languages (or English with fallback accepted by the owner)	int.notify.manage (MFA for customer channels)	NOTIFY_RULE_ACTIVATED
ACTIVE	RETIRED	RetireRule	reason	int.notify.manage	NOTIFY_RULE_RETIRED

4.4 E-invoice submission (readiness)

From	To	Trigger / command	Guard	Permission	Audit
NOT_SUBMITTED	PENDING	invoice.issued.v1 with e-invoicing enabled for the entity	adapter configured (stub in phase 1)	system	EINVOICE_PENDING
PENDING	SUBMITTED / ACCEPTED / REJECTED	adapter response (phase 3)	–	system	EINVOICE_*

5. Contract

5.1 Commands

Command	Input (essentials)	Permission	Events
RegisterApiClient / RotateCredential / AmendScopes / SetRateLimit / SuspendClient / ReinstateClient / RevokeClient	name, kind, credential kind, scopes; reason	int.client.manage (MFA)	api_client.*.v1
RequestJournalExport / AcknowledgeJournalExport / RegenerateJournalExport	entity, period, format, provisional?; export id	int.journal.export / ack	journal.requested/generated/failed/acknowledged/superseded.v1
RegisterPostingMap (internal, from M5/M7 seeds)	doc type, line kinds, roles	system	posting_map.registered.v1
DefineTemplate / ActivateTemplate / RetireTemplate	channel, three languages, placeholders	int.notify.manage	notification_template.*.v1
DefineRule / ActivateRule / RetireRule	event type, predicate, template, audience, channels	int.notify.manage (MFA for customer channels)	notification_rule.changed.v1
ConfigureProvider / TestProvider	channel, providers, sender id, cap	int.provider.manage (F, MFA)	provider.configured.v1
EnableEInvoicing (entity) / SubmitEInvoice (stub)	entity; invoice	int.einvoice.manage (F)	einvoice.*.v1
PublishApiBundle (doc 33)	version, slices	int.api.publish (F)	api_bundle.published.v1

5.2 Queries (admin) and the public operations

Query / operation	Returns	Scope rule
ListApiClient(entity) / GetClientUsage(client)	Clients, scopes, call statistics	OWN admin
ListJournalExports(entity) / GetJournalExport(id) / GetJournalLines(id)	Exports and lines; download link	OWN accounts; API journal:read
ListNotificationRules / Templates / GetNotificationLog(scope)	Rules, templates, delivery log (recipient hashed)	OWN; F for federation rules
GetPostingMap(doc type)	Roles by line kind	Any authenticated
Public: GET /v1/trading/invoices, credit-notes, debit-notes, payment-receipts, grns; GET /v1/inventory/.../balances; GET /v1/trading/relationships/{id}/exposure and /statement; GET /v1/integration/journal-exports; GET /v1/customers/... (customers:read)	The module queries' public subsets through their own controllers	Client's entity: OWN or PARTY per the module's RLS
Public: POST /v1/integration/journal-exports/{id}/acknowledge	The one write	journal:ack

5.3 Events published

Event	Payload	Consumers
api_client.registered/rotated/scopes_changed /suspended/reinstated/revoked.v1	client id, entity, scopes	Identity provider mapper (client scopes), audit, M8
journal.requested/generated/failed/acknowledged/superseded.v1	export id, entity, period, totals, object key	Notifications (accounts role), M8
notification_rule.changed.v1 / notification_template.*.v1 / provider.configured.v1	rule id, event type, status	Kernel dispatcher cache, audit
einvoice.pending/submitted/accepted/rejected.v1	invoice id, status, provider ref	M4 (e_invoice_status metadata), M8
api_bundle.published.v1	version, deprecations	Documentation site, audit

6. Key flows

Flow	Sequence	Failure paths
6.1 Onboarding the accounting system	MPCS administrator registers an API client (kind ACCOUNTING) with scopes journal:read, journal:ack, documents:read (MFA) → credential issued once, shown once → the consumer authenticates with client credentials → token carries the entity and scopes → reads invoices and the month's journal export → acknowledges	A scope beyond the administrator's own permissions is refused; a suspended client gets 403; rate limit 429
6.2 Month-end journal export	Accounts requests the export for September (all locations' day-close past 30 September) → JournalBuilder walks INV, CN, DN2, PRC (seller), GRN and INV (buyer), WOF, ADJ, CPR with their maps → lines by role; totals balance; hash; JSON and CSV stored → GENERATED → notification to accounts → consumer downloads and acknowledges	A document without a map for its line kind: FAILED with an ALERT naming it; a provisional export (period not closed) is marked PROVISIONAL in the file and superseded later
6.3 A late document	An invoice for September issued on 3 October after the export → not edited into the export; a supplementary export for the same period lists only documents issued since the last export's generated_at; the consumer posts the supplement	–
6.4 Dispatch alert	M4 delivery_note.dispatched.v1 → kernel dispatcher matches the federation rule (audience: stores role at the buyer; channels SMS, IN_APP) → template rendered in the recipient's language → HttpSmsAdapter sends via the primary provider → log	Primary provider down: secondary after the third attempt; quiet hours: deferred to 07:00; no consent needed (staff channel)
6.5 Customer statement SMS	M7 customer_statement.issued.v1 → rule with audience CUSTOMER → consent STATEMENTS checked → 160-character summary in the customer's language → sent	No consent: SUPPRESSED in the log
6.6 E-invoicing enabled later	Federation enables e-invoicing for an entity; invoice.issued.v1 creates PENDING submissions; the stub records; when the IRD adapter lands (phase 3) it drains PENDING and posts ACCEPTED/REJECTED as metadata on M4's invoice	–
6.7 Publishing the API bundle (doc 33)	At cross-module reconciliation: collect every module's OpenAPI slice, validate consistency (shared schemas, error types, security scheme), mark public operations, generate the documentation site and the typed clients → version 1.0.0 published	A slice that disagrees on a shared schema fails the build

7. Configuration

Item	Default	Scope	Who may change
integration.client_rate_limit_per_minute	600	Federation; per client override	int.client.manage
integration.credential_overlap_days	7	Federation	–
integration.journal_formats	[JSON, CSV]	Federation	int.journal.export
integration.journal_unacknowledged_review_days	30	Federation	–
notify.quiet_hours (SMS)	22:00–07:00	Entity	int.notify.manage
notify.daily_cap_per_channel	provider-dependent	Federation	int.provider.manage
notify.kill_switch (channel)	off	Entity	int.notify.manage
einvoice.enabled	false	Entity (F sets)	int.einvoice.manage
api.deprecation_notice_months	12	Federation	–

8. Screens as paper

Screen	Paper mirrored	Surface	Roles	Notes
API clients	Access register	W	Entity admin (MFA)	Register, scopes, rotate (credential shown once), suspend, usage
Journal exports	Journal	W	Accounts	Period, format, provisional flag, totals, download, acknowledgement state, supplements
Posting map viewer	Chart mapping	W	Accounts, auditor	Roles by document type and line kind (read-only)
Notification templates	Notice text	W	Federation, MPCS	Three languages, placeholders, SMS length meter, preview

Screen	Paper mirrored	Surface	Roles	Notes
Notification rules	Notice list	W	Federation, MPCs	Event, audience, channels; activation
Notification log	Dispatch book	W	Entity	Status, attempts, suppression reasons; recipients hashed
Providers	–	W	Federation (MFA)	Primary and secondary per channel; test send
E-invoicing	–	W	Federation	Enable per entity; submission states (stub)
API documentation	–	W (public)	Clients	Generated from the bundle; versions and deprecations

9. Non-functional design

Aspect	Design
9.1 Security	Clients are principals under the same scope filter and RLS; scopes never exceed the granting administrator's permissions; credentials hashed (secrets) or certificate-bound; rotation with overlap; rate limits at the edge and in-process; every write audited; the public subset is explicitly marked in the bundle.
9.2 Isolation	Adapters run in the worker profile with timeouts and circuit breakers; a provider outage never blocks a business transaction (notifications are asynchronous by design).
9.3 Correctness	Journal exports balance by construction; a map defect is an ALERT; exports are immutable and superseded, never edited; hashes let the consumer verify a download.
9.4 Privacy	Notification logs hash recipients and keep no bodies; customer channels require consent; the public API exposes customer data only under customers:read to the owning MPCs.
9.5 Compatibility	Additive changes within /v1; 12-month deprecation; generated clients versioned with the bundle.

10. Test strategy

Layer	What is tested
Unit	Scope containment on registration; rotation overlap; JournalBuilder line derivation for every mapped document type and side; balance assertion; supplement selection by generated_at; template rendering and SMS length; audience resolution incl. consent
Contract	Every admin operation; every public operation under a client token (OWN and PARTY masking); rate limit 429; idempotent acknowledgement
State machine	Clients, exports, rules, submissions
Property-based	Random document sets: every export balances; supplement plus original equals a fresh full export
Adapters	SMS and e-mail adapters against local fakes (compose: mailpit; an HTTP SMS stub); failover; quiet hours
Scenario	Flows 6.1–6.7 incl. the accounting consumer simulator and the bundle publication
Bundle	All module slices merge without conflicts; shared schemas identical; generated clients compile

11. Decision requests

#	Point	Proposed	Decider
DR-1	Public write operations in phase 1	Only journal acknowledgement	Federation
DR-2	Journal export formats before the accounting package is chosen	Generic JSON and CSV with account roles; package adapter later	Federation accounts
DR-3	Which document types carry posting maps in phase 1	INV, CN, DN2, PRC (seller); GRN, INV (buyer); WOF, ADJ, RPK (M5); CPR (M7)	Accountant review (doc 20 M9 risk)
DR-4	SMS provider(s)	Procurement item; adapter is provider-neutral; two providers for failover	Federation procurement
DR-5	MPCS-authored notification rules	Allowed within their scope for their own events	Federation
DR-6	Webhooks	Phase 2	Federation
DR-7	Integration tables (CR-29-1 to doc 18)	Adopt	Architecture review

12. Traceability

Section	Identifiers
3.1–3.2	R-07; doc 17 §10.1; doc 19 §2, §3; doc 33
3.3	FR-BIL-140; J-02; ADR-01, ADR-05; doc 24 §3.9; doc 25, 27 posting maps
3.4	R-08; doc 19 §7; M7 consent
3.5	FR-BIL-110; J-04; R-06; doc 24 §3.6
3.6–3.7	M-10; doc 17 §10.2 webhooks
9	NFR-SEC; ADR-27

End of document. Draft for design review; the contract freezes with the doc 33 bundle at cross-module reconciliation, targeted Stage 2 week 7. Implementation guide: 29A.